OMEX Combination Report — SLS_Concentration=2000.0

vcornea_sls_sweep_combo002 (point 2 of 3) | 5 replicate archive(s) | vivarium_wrapper 1.0.0

Combination Overview

Scan point	SLS_Concentration=2000.0 (point 2 of 3)
Simulation	SimTime 30 days (= 7200 MCS = 720 hours) · injury=yes (chemical)
Replicates	5 (this report aggregates 5 successful archive(s); plots show each replicate plus the across-replicate mean)

How Replicates Appear in the OMEX

Each replicate is packaged as its OWN per-run .omex (under *replicate_N/*), carrying only that replicate's data and a *replicate_id* / *total_replicates* in run_metadata.json. The per-replicate SED-ML declares a replicate *repeatedTask* over a uniformRange (1..5) [present] — i.e. it documents the N-replicate design even though each archive holds one replicate. This report reads all replicate archives of the combination and aggregates them.

Replicate	run_metadata	built-in SED-ML	manifest	biosimulators
replicate_1	id 1 of 5	PASS	PASS	PASS (0 errors, 1 warning group)
replicate_2	id 2 of 5	PASS	PASS	PASS (0 errors, 1 warning group)
replicate_3	id 3 of 5	PASS	PASS	PASS (0 errors, 1 warning group)
replicate_4	id 4 of 5	PASS	PASS	PASS (0 errors, 1 warning group)
replicate_5	id 5 of 5	PASS	PASS	PASS (0 errors, 1 warning group)

Provenance (OMEX Metadata)

Field	Value
Creator (experimenter)	Joel Vanin
Created	2026-10-21T11:54:10
Archived	2026-10-21T18:38:01
License	https://spdx.org/licenses/CC-BY-4.0.html
Citation	Vanin et al. (2025). V-Cornea: a computational model of the corneal epithelium.
Keywords	CompuCell3D, cornea, epithelium, agent-based model, wound healing

From metadata.rdf (COMBINE OMEX-Metadata). The creator is attribution/provenance, not authentication.

Run Identity (stochastic realization & code)

Replicate	RNG seed	Origin	Schema fingerprint	Code
replicate_1	1574844843	drawn	sha256:d3c5ca23a40e24f9	build_stamp
replicate_2	643665948	drawn	sha256:d3c5ca23a40e24f9	build_stamp
replicate_3	1797272279	drawn	sha256:d3c5ca23a40e24f9	build_stamp
replicate_4	321534142	drawn	sha256:d3c5ca23a40e24f9	build_stamp
replicate_5	346823061	drawn	sha256:d3c5ca23a40e24f9	build_stamp

Replicates of a combination share a parameterization and differ only in the random number generator seed, so a difference between them is stochastic rather than configured. Seeds marked *drawn* were generated independently at launch and recorded afterwards; seeds marked *explicit* were set by the operator (a pinned seed applies to the first replicate only). The schema fingerprint identifies the parameter definitions in force when the run was configured, and the code entry the source revision that executed it.

Parameter Credibility (run values against the schema's documented evidence)

79 parameters, 1 changed from default; 27 within documented evidence, 0 extrapolations, 0 inadmissible, 0 numeric without a documented range, 21 switches/categories.

Parameter	Run value	Default	Unit	Credibility	Evidence for the range assessed

Initial Chemical Concentration (SLS_Concentration)	2000	750	Arbitrary Units	admissible (no physiological range documented)	calibration: Scott L et al. (2010), Maurer JK et al. (2002), Maurer JK et al. (1997); S10 Table, p. 19

Credibility places each executed value against the three range tiers the schema documents: the implementation-admissible range (outside = inadmissible), the evidence-supported range (outside = extrapolation beyond the documented evidence) and the recommended range for this model's context of use. The evidence column names the basis and citation the schema records for the supported range; *not recorded* means the bound exists but its justification has not been documented, which is itself an audit finding. Numeric parameters without any documented range are listed as unassessable; switches and categories are not range-assessed. Full detail for every parameter is in parameter_audit.json inside the archive.

Reading the SED-ML (representative replicate)

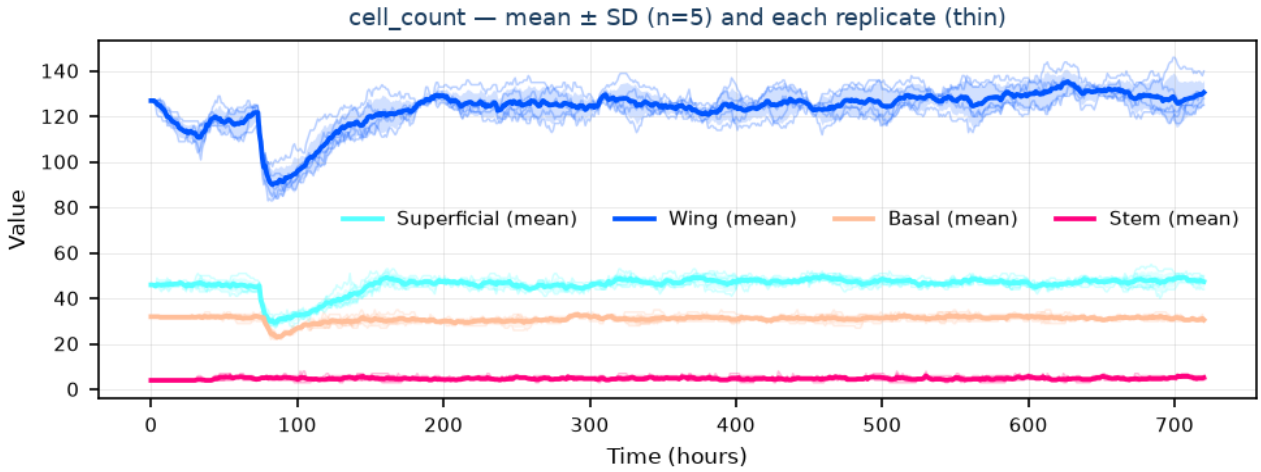
What the file states	Conclusion
Model model/vCornea_v2.cc3d · urn:sedml:language:compucell3d	Agent-based CC3D model, **bundled in the archive** so model/@source resolves (self-contained); still descriptive, not re-executable by a SED-ML engine.
Simulation uniformTimeCourse 0→7200, 7200 steps, KiSAO KISAO:0000241	30-day run = 720 h = 7200 MCS (time course is in MCS).
Replicates repeatedTask present, total_replicates=5	Stochastic replicates; means/spread across replicates are meaningful (this report uses 5).
DataGenerators: 11; dataDescriptions: 3	Signals the SED-ML indexes; thickness parquet is packaged but not indexed.

Recorded Outputs (DataDescriptions → columns)

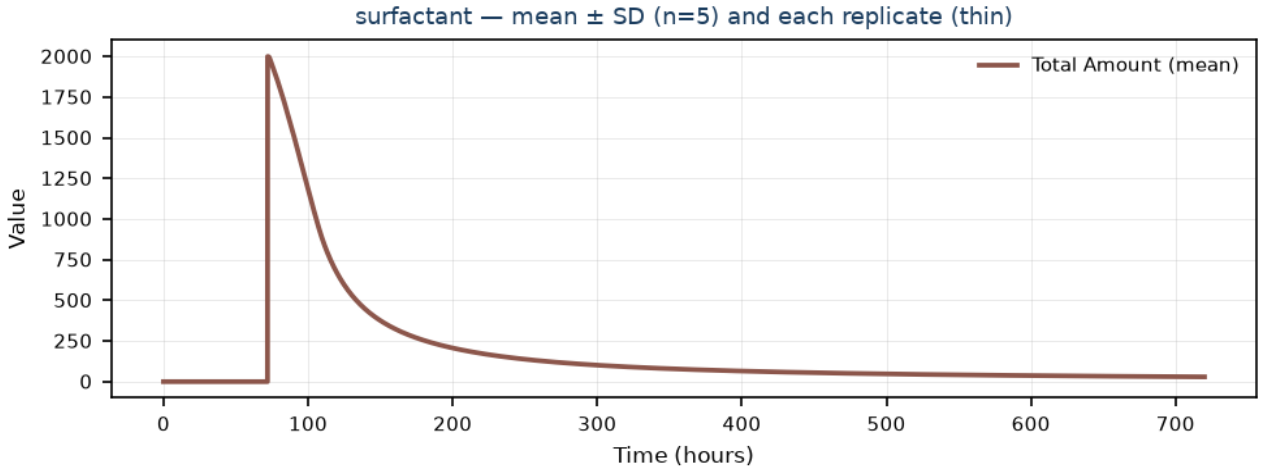
Source file	Format	Columns
cell_count_7201.csv	csv	Superficial, Wing, Basal, Stem, Time
surfactant_7201.csv	csv	Total Amount, Time
thickness_rep_7201.parquet	//purl.org/NET/mediatypes/application/x-parquet	Time, Bin, Height

Results (replicate-aware, from the packaged data)

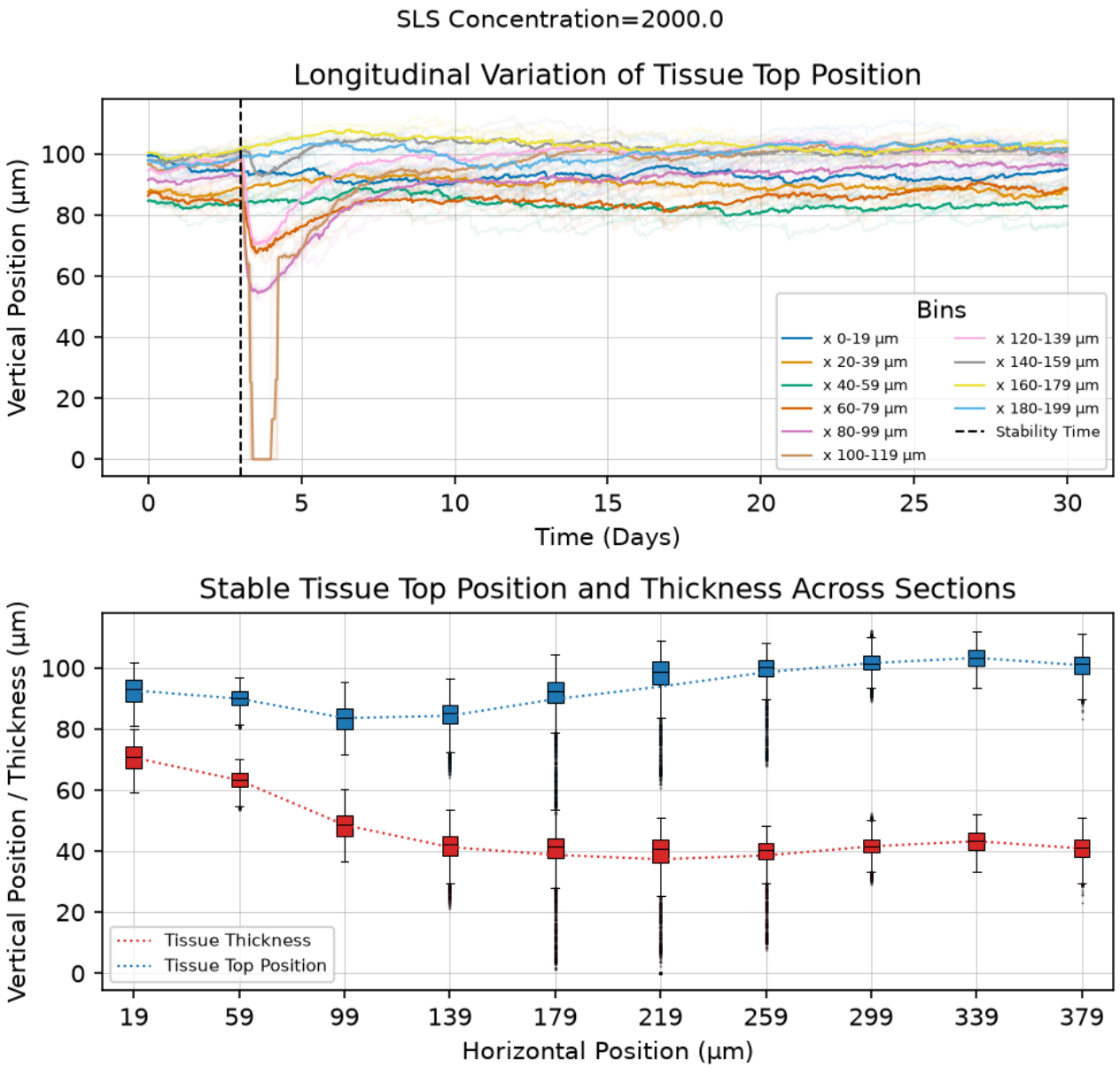
Plotted from the CSV/parquet inside the 5 replicate archive(s). Thin lines are individual replicates; the bold line is the across-replicate mean and the shaded band ±1 SD across replicates. Time is shown in hours (as stored in the data).



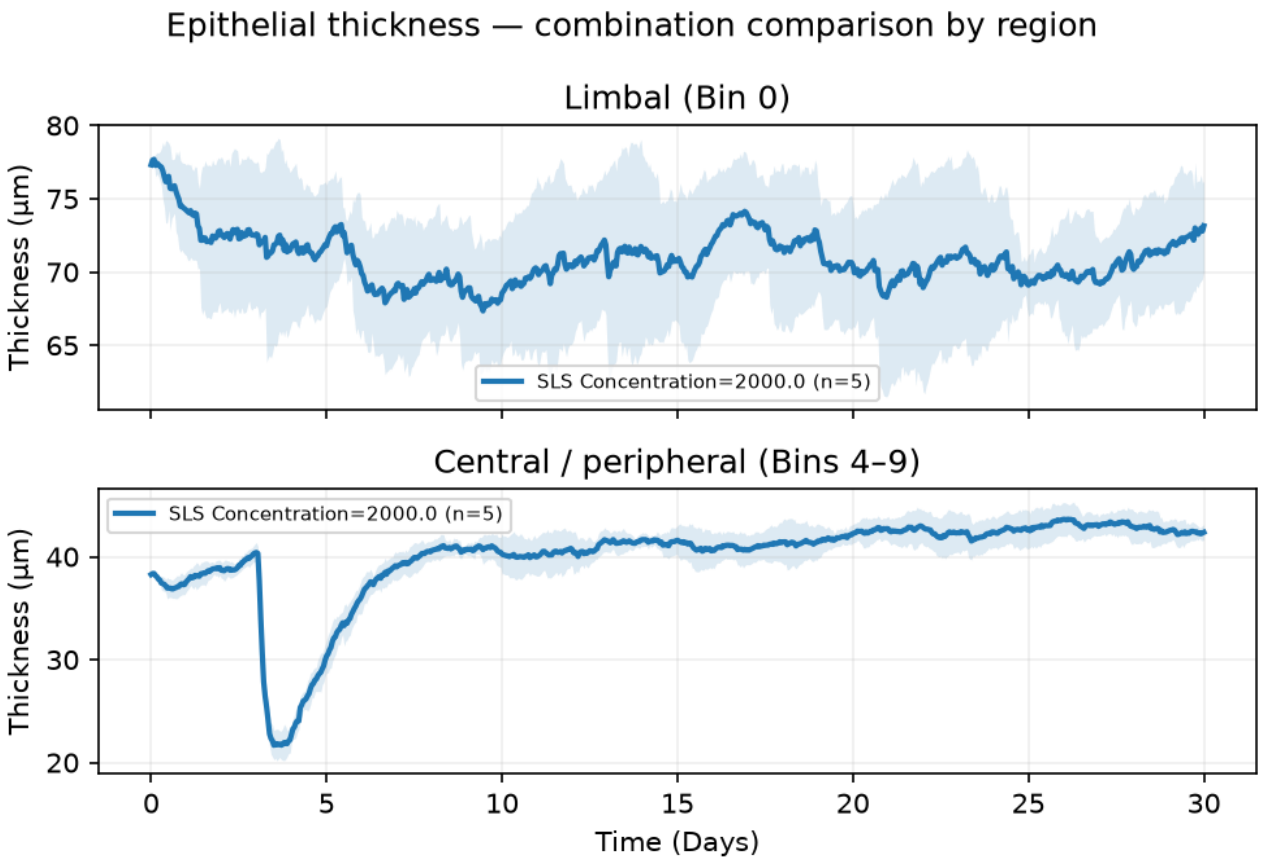
cell_count: Superficial, Wing, Basal, Stem vs Time — n = 5 replicate(s); bold = across-replicate mean, shaded = ±1 SD across replicates, thin = individual replicates



surfactant: Total Amount vs Time — n = 5 replicate(s); bold = across-replicate mean, shaded = ±1 SD across replicates, thin = individual replicates



Tissue thickness, drawn by the V-Cornea GUI's own analysis (Analyze Results → Thickness, paper view; Vanin et al. 2025, Fig 8). Top: tissue top position (vertical centre of mass of the top cells) in each of the 10 X-bins over time, as the model records it; faint lines are the 5 replicate(s), solid lines their mean. Bottom: from the stability time (day 3) on, top position (blue) and thickness (red) across X; thickness = (top position – membrane baseline) × 2 µm/voxel.



Epithelial thickness in µm over time in the limbus (bin 0) and the central/peripheral cornea (bins 4–9): mean of the 5 replicate(s), shaded ±1 SD across replicates (Analyze Results → Thickness, comparison view).

Archive Contents (per replicate)

File	Size (B)	Role
Epithelium.piff	692,401	—
Epithelium_new_5545mcs.piff	656,912	—
Parameters.py	2,270	CC3D parameter file applied to this run
cell_count_7201.csv	13,598	—
environment.json	1,292	—
manifest.xml	2,519	OMEX manifest (COMBINE archive index)
metadata.rdf	1,657	OMEX Metadata (provenance: creator, license, citation)
model/Epithelium.piff	692,401	Bundled CC3D model file (so model/@source resolves)
model/Epithelium_new_5545mcs.piff	656,912	Bundled CC3D model file (so model/@source resolves)
model/Simulation/Parameters.py	2,270	CC3D parameter file applied to this run
model/Simulation/vCornea_v2.py	1,205	Bundled CC3D model file (so model/@source resolves)
model/Simulation/vCornea_v2.xml	13,788	Bundled CC3D model file (so model/@source resolves)
model/Simulation/vCornea_v2Steppables.py	67,455	Bundled CC3D model file (so model/@source resolves)
model/vCornea_v2.cc3d	267	Bundled CC3D model file (so model/@source resolves)
parameter_audit.json	256,577	—
run_metadata.json	3,563	Authoritative parameter/run provenance
simulation.sedml	12,509	SED-ML L1V4 simulation description
status.json	492	—
stderr.log	422	CC3D run log (stderr)
stdout.log	1,421	CC3D run log (stdout)

surfactant_7201.csv	169,883	—
thickness_rep_7201.parquet	76,733	—
vCornea_v2.cc3d	267	—

Each of the 5 replicate archives has this same layout.